

Skeletal muscle proteomic signature and metabolic impairment in pulmonary hypertension

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Abstract

Exercise limitation comes from a close interaction between cardiovascular and skeletal muscle impairments. To better understand the implication of possible peripheral oxidative metabolism dysfunction, we studied the proteomic signature of skeletal muscle in pulmonary arterial hypertension (PAH). Eight idiopathic PAH patients and eight matched healthy sedentary subjects were evaluated for exercise capacity, skeletal muscle proteomic profile, metabolism, and mitochondrial function. Skeletal muscle proteins were extracted, and fractionated peptides were tagged using an iTRAQ protocol. Proteomic analyses have documented a total of 9 downregulated proteins in PAH skeletal muscles and 10 upregulated proteins compared to healthy subjects. Most of the downregulated proteins were related to mitochondrial structure and function. Focusing on skeletal muscle metabolism and mitochondrial health, PAH patients presented a decreased expression of oxidative enzymes (pyruvate dehydrogenase, $p < 0.01$) and an increased expression of glycolytic enzymes (lactate

dehydrogenase activity, $p < 0.05$). These findings were supported by abnormal mitochondrial morphology on electronic microscopy, lower citrate synthase activity ($p < 0.01$) and lower expression of the transcription factor A of the mitochondria ($p < 0.05$), confirming a more glycolytic metabolism in PAH skeletal muscles. We provide evidences that impaired mitochondrial and metabolic functions found in the lungs and the right ventricle are also present in skeletal muscles of patients.

Key message

- Proteomic and metabolic analysis show abnormal oxidative metabolism in PAH skeletal muscle.
- EM of PAH patients reveals abnormal mitochondrial structure and distribution.
- Abnormal mitochondrial health and function contribute to exercise impairments of PAH.
- PAH may be considered a vascular affliction of heart and lungs with major impact on peripheral muscles.

Keywords Pulmonary hypertension · Skeletal muscle · Mitochondrial metabolism · Proteomic · Exercise capacity

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Introduction

Pulmonary arterial hypertension (PAH) remains to date an enigmatic disease. It is characterized by a pulmonary pathological vascular remodeling syndrome and right heart failure [1]. Despite the state-of-the-art modern therapies, PAH remains with important mortality and morbidity, poor exercise capacity, and health-related quality of life [2–4].

In addition to an increase in cardiac output, physical exercise also requires a global and integrated response of many systems, including the respiratory system for oxygen uptake and carbon dioxide clearance and peripheral vessels to

increase muscle blood flow and deliver oxygen to generate energy and external work. Not surprisingly, in addition to pulmonary hemodynamic impairment, exercise capacity in PAH correlates with impaired quadriceps muscle function [5] and vascularization [6]. Nonetheless, the mechanisms of exercise intolerance in PAH remain poorly understood.

A growing body of evidence establishes that an impaired cellular metabolism is central in PAH pathophysiology [7]. Our laboratory and others have shown that mitochondria account to this metabolic dysfunction through mitochondrial hyperpolarization, endoplasmic reticulum stress, and enhanced anaerobic glycolysis over glucose oxidation and contribute to the proliferative/anti-apoptosis phenotype of PAH pulmonary artery smooth muscle cells [8–10]. Mitochondrial-induced vascular remodeling resulting to a metabolic switch toward glycolysis has also been described in PAH right ventricle (RV) [11] and shown to contribute to the progression toward RV failure [12]. Interestingly, it has been recently showed that skeletal muscle mitochondrial dysfunction precedes RV impairment in an animal model of PAH [13]. Altogether, these recent data suggest that mitochondrial abnormalities observed in the lungs and RV of PAH patients could also be present in their skeletal muscles. As such, skeletal muscle mitochondrial abnormalities may contribute to peripheral muscle dysfunction and exercise intolerance in PAH.

Therefore, the aim of this study was to assess whether mitochondrial abnormalities were also observed in skeletal muscles of PAH patients. The investigation of specific abnormal mitochondrial pathways and interactions remains narrow, complex, and often left to chance. To overcome this limitation, our first objective was to map the proteomic structure of PAH patients' skeletal muscle, allowing us a broader view of the potent molecular abnormalities. Based on these proteomic results, our second objective was to confirm this proteomic signature by verifying the abnormalities in mitochondrial structure and metabolic function of PAH skeletal muscles. We hypothesized that an abnormal proteomic structure of PAH skeletal muscle was principally affecting the mitochondria, which would translate to a metabolic switch toward glycolysis as observed in PAH RV and lungs. We also hypothesized that skeletal muscle oxidative capacity would correlate with PAH patients' exercise capacity.

Methods

Subjects

Eight patients with modified New York Heart Association (NYHA) functional class II–III idiopathic PAH with recent hemodynamic assessment were recruited at the Institut Universitaire de Cardiologie et de Pneumologie de Québec (IUCPQ). PAH diagnosis and management were made in

accordance to the most recent guidelines [1]. Eight healthy controls were matched for age, sex, and body mass index. Experiments were made in accordance with the Université Laval and the IUCPQ biosafety and ethics committee (CÉR-20643). All patients gave informed consent before the study initiation.

Study design

Exercise capacity, dominant quadriceps voluntary strength and endurance, and mid-thigh, non-dominant leg percutaneous muscle biopsy of the *vastus lateralis* were made for all participants, as routinely performed in our laboratory (see the online supplemental method for more details) [5]. *Vastus lateralis* morphology was assessed for all participants, whereas *vastus lateralis* metabolic and mitochondrial assessment was made according to tissue availability. Out of our cohort, fresh biopsies from four PAH patients and four matched controls were sent for proteomic analysis (Proteomics Center, Quebec City (<http://proteomics.crchul.ulaval.ca/en/>)) and electronic microscopy analysis (Institut de Biologie Intégratives et des Systèmes (IBIS), Université Laval). Biopsy specimens were stored at the IUCPQ site of the Tissue Bank of the FRQS (http://swrsr.crc.chus.qc.ca/en/tissue_bank.asp), excepted for specimens sent to proteomic analysis.

Peripheral muscle morphology

Muscle sections were stained to see myofibrillar adenosine triphosphatase activity according to the single-step modified ethanol technique, as routinely performed in our laboratory [14] (see the online supplemental method for more details).

Proteomics analysis

After harvest, specimens from four PAH patients and four matched healthy controls were rinsed in PBS, flash-freeze, and stored at -80°C . Protein extraction was performed on 5 to 20 mg of frozen tissues in liquid-nitrogen-chilled microcentrifuge tube and pestle using a buffer of sodium deoxycholate (Sigma) with proteases and phosphatase inhibitors. Samples were centrifuged and a Bradford assay was performed on the supernatant for protein levels. One hundred micrograms of proteins for each sample were sent to the proteomic platform for iTRAQ labeling and mass spectrometry analysis.

iTRAQ sample labeling Each sample of protein was used for iTRAQ labeling. Proteins were then reduced and alkylated according to the iTRAQ kit manufacturer's instruction (ABSciex). Samples were digested with trypsin (Sequence grade Modified, Promega) using 1:20 ratio overnight at

37 °C. After digestion, peptides were acidified to precipitate with deoxycholate and then purified with an oasis HLB cartridge (1 cc, 10 mg, Water Corp.) and lyophilized. Dried peptides were dissolved in 30 µl 0.5 M TEAB and labeled with iTRAQ label reagent (ABSciex). Four-plex labeling was performed for 2 h at room temperature in the dark. Labeled peptides were combined in one tube and dried using a SpeedVac. Samples were then cleaned up using HLB cartridge (Water Corp.), dried and reconstituted in 200 µl HPLC water and 1/100 ampholytes pH 3–10 (Bio-Rad). Then, peptides were fractionated with 7-cm immobilized pH gradient strips (pH 3–10), using an isoelectric focusing method, and performed for 10,000 Vh. Immobilized pH gradient strips were cut in 14 fractions, and peptides were extracted in 2 % ACN-1.0 % FA solution followed by 50 % ACN-1 % FA. Finally, fractions were dried with the SpeedVac.

Mass spectrometric analysis For LC-MS/MS analysis of iTRAQ labeled samples, each isoelectric focusing fraction was resuspended in 20 µl of 0.1 % formic acid. Mass spectrometry analysis was performed on a TripleTOF 5600 mass spectrometer fitted with a nanospray III ion source (ABSciex, Concord, ON) and coupled to an Agilent 1200 HPLC (Agilent, California). Two-microliter samples from a 1/20 dilution of the 20 µl were injected by the Agilent 1200 autosampler onto a trapping column (Zorbax 300SB-C18) for desalting then onto a self-packed PicoFrit column (New Objective) packed with an isopropanol slurry of 5-µm Jupiter C18 (Phenomenex) stationary phase using a pressure vessel (Proxeon) set at 700 psi. The length of the column was 15 cm. Samples were run using a 90-min gradient from 5 to 35 % solvents (solvent A 0.1 % formic acid in water; solvent B 0.1 % formic acid in acetonitrile) at a flow rate of 300 nl.min⁻¹. Data was acquired using an ion spray voltage of 2.4 kV, curtain gas of 30 psi, nebulizer gas of 8 psi, and an interface heater temperature of 125 °C. An information-dependant acquisition (IDA) method was set up with the MS survey range between 400 and 1250 amu (250 ms) followed by dependent MS/MS scans with a mass range set between 100 and 1800 amu (50 ms) of the 20 most intense ions in the high sensitivity mode with a 2+ to 5+ charge state. Dynamic exclusion was set for a period of 12 s and a tolerance of 100 ppm. Rolling collision energy was used and iTRAQ reagent collision energy adjustment was on.

Metabolic activity

From our initial cohort and according to tissue availability, muscle metabolic enzymatic activity was tested in seven PAH patients and in six matched controls. Pyruvate dehydrogenase (PDH) enzyme activity microplate assay kit (Abcam, ab109882) and lactate dehydrogenase (LDH) enzyme activity assay kit (Abcam,

ab102526) were used according to the manufacturer's instructions (Abcam Inc., Toronto, Canada). For each enzyme, protein concentrations were determined prior to detergent addition using Bio-Rad Protein Assay Dye Reagent Concentration (Bio-Rad, #500-0006). Mitochondrial density and integrity were monitored by quantifying citrate synthase (CS) activity [15] and by the measurement of transcription factor A of the mitochondria (TFAM), respectively. Briefly, TFAM was obtained by Western blots. Proteins were electrotransferred into PVDF membranes, subsequently washed, and incubated at 4 °C with primary antibody TFAM (Millipore ABE483). After incubation, membranes were washed in Tris-buffered saline (TBS)-Tween 20 (TBS-T) and incubated using horseradish peroxidase (HRP)-conjugated secondary antibody (Promega) at 4 °C for 1 h. Protein bands were revealed using Chemidoc (Bio-Rad). The bands were quantified by ImageJ software (version 1.46) and normalized to actin, which served as an internal control.

Table 1 Clinical characteristics of the population study

Clinical variable	Healthy controls	PAH	<i>p</i> value
<i>N</i>	8	8	
Demographics			
Age (year)	42 (11)	41 (10)	0.84
Females/males	8/0	8/0	
BMI (kg m ⁻²)	24.9 (2.2)	26.0 (4.9)	0.62
PAH type			
Idiopathic	NA	8	
Modified NYHA FC			
II	NA	6	
III	NA	2	
Pulmonary hemodynamics			
mPAP (mmHg)	NA	50 (11)	
PCWP (mmHg)	NA	11 (3)	
CI (l min ⁻¹ m ⁻²)	NA	2.9 (0.9)	
PVR (dynes s cm ⁻⁵)	NA	620 (307)	
Exercise capacity and quadriceps strength			
6MWT (m)	NA	449 (35)	
6MWT predicted (%)	NA	75 (12)	
VO ₂ max (ml O ₂ min ⁻¹ kg ⁻¹)	26.2 (1.7)	13.9 (1.4)	<0.0001
VO ₂ max predicted (%)	112 (24)	59 (24)	<0.01
Voluntary strength (kg)	31.6 (3.7)	24.2 (4.0)	<0.01
Quadriceps endurance (N m ⁻¹)	1906 (487)	1377 (302)	0.03

Data are presented as mean (standard deviation)

BMI body mass index, *NYHA FC* New York Heart Association Functional Class, *mPAP* mean pulmonary artery pressure, *PCWP* pulmonary capillary wedge pressure, *CI* cardiac index, *PVR* pulmonary vascular resistance, *6MWT* 6-min walk test, *VO₂* oxygen uptake

Bioinformatics and statistical analysis

Proteomic data files were submitted for simultaneous searches using Protein Pilot version 4.5 software (ABSciex) using the Paragon and Progroup algorithms (Shilov) and the integrated false discovery rate analysis function (Tang). Protein Pilot was set up to search the uniprot complete proteome human proteins database (84,848 sequences) with MMTS as a fixed modification on cysteine. Variable peptide modifications included methionine (M) oxidation and iTRAQ labeling of the N-terminal, lysine (K), and tyrosine (Y). Automatic normalization of quantitative data (bias correction) was performed to correct any experimental or systematic bias. The detected protein threshold (unused protscore) in the software was set to achieve a 1 % false discovery rate.

Differentiated proteins were selected based on criteria related to identification confidence (false discovery rate <1 %, with at least one peptide identified with Protein Pilot confidence score >95 %), quantification confidence (p value <0.05 using a two-sided Fisher's test), fold change amplitude (absolute value of fold change of <0.80 for downregulated proteins and >1.2 for upregulated proteins), and prevalence in replicates (proteins must meet the previous criteria in all four replicates). Gene Ontology (GO) annotation enrichment was calculated using GoMineR [16, 17] and BinGO (version 3.0.2) and plotted using Cytoscape (version 3.1.1) [18]. The

background against which the enrichments was calculated was composed of 1600 human genes associated with a protein identified with false discovery rate <1 % in the mass spectrometry experiments. Further statistical analyses on the proteomics data (clustering, PCAs, heatmaps) were done on the R statistical platform (version 3.0.1). The heatmap was obtained through the “heatmap.2” function of the *gplots* package, whereas the clusters were based on the clustering function *hclust* and the distance matrix function *dist*.

Statistical analysis for muscle typology, PDH, LDH, CS, and TFAM was made using unpaired Student's t test. Because the maximal oxygen uptake is representative of the maximal oxidative capacity, correlations between the maximal oxygen uptake (VO_2 max) and markers of muscle oxidative capacity (PDH, CS, and TFAM) were assessed using Pearson's r . Data are presented as mean $\pm$ standard deviation (SD) unless otherwise specified. Metabolic data statistics were made using GraphPad Prism 6.0e.

Results

Baseline characteristics of the patient and control subjects are shown in Table 1. PAH patients were mostly in NYHA

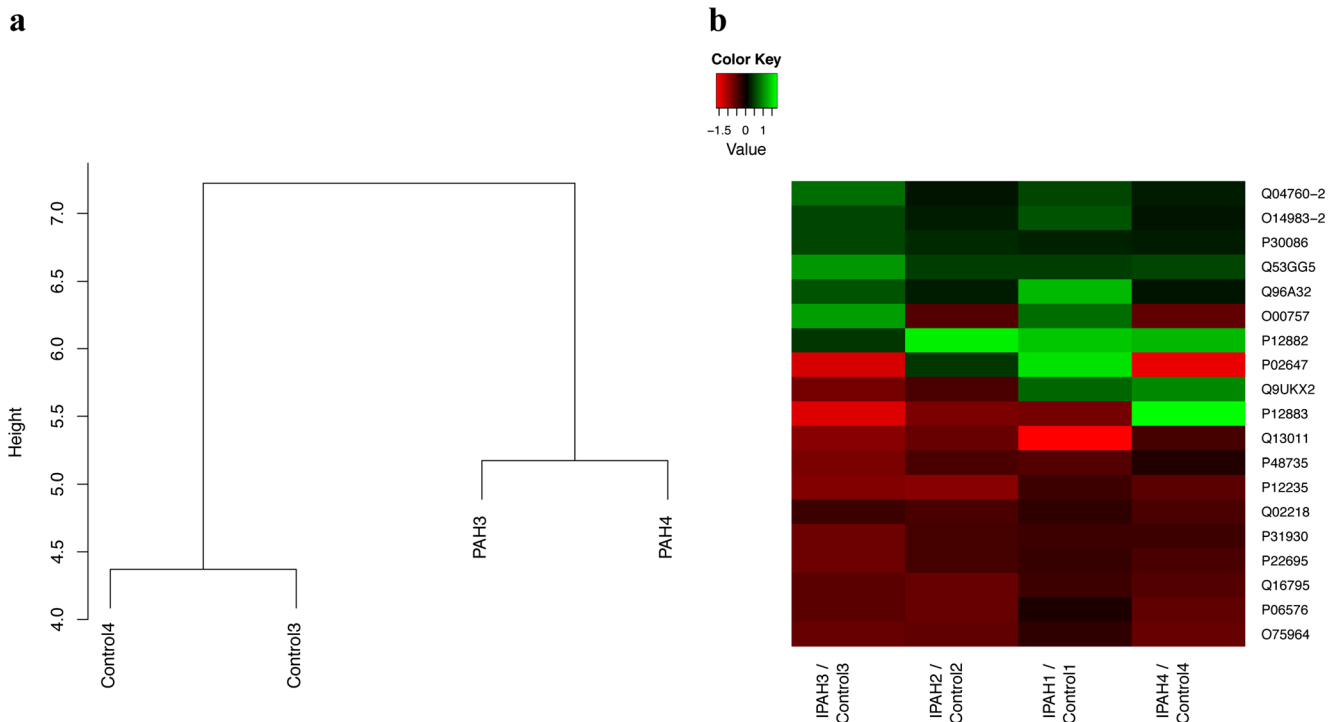


Fig. 1 Proteomic signature of PAH skeletal muscle. **a** Schematic representation of clusterization based on 231 proteins displaying that iPAH and control proteomic profile are statistically regrouped into two different families. **b** Heatmap generated from proteomic analysis of 4 controls and 4 idiopathic PAH patients represents the ratio of PAH/control

protein expression of the 19 proteins which are significantly upregulated (green) or downregulated (red) in PAH skeletal muscle compared to healthy control. Note that color intensity is proportional to fold of change of the ratio (color figure online) and numbers represent the Uniprot code numerization

functional class 2 but displayed significant decrease in exercise capacity and skeletal muscle function.

Abnormal skeletal muscle proteomic signature in PAH

Using clustering analysis based on 231 matched proteins, we showed that skeletal muscle protein expression of PAH and matched controls could be grouped into two different families (Figs. 1a, E1, and E2, online supplement). Heatmap generated from proteomic analysis displayed 9 downregulated proteins (red) and 10 upregulated proteins (green) in our PAH patient samples compared to matched controls, suggesting impairments of skeletal muscle protein expression in PAH (Fig. 1b). Proteins for which the expression was strongly and significantly different in PAH are listed in Table 2. We first observed an important decrease of key proteins from mitochondrial oxidative phosphorylation *complexes I* (NADH dehydrogenase [ubiquinone] 1 alpha

subcomplex subunit 9), *complex III* (cytochrome b-c1 complex subunit 1, cytochrome b-c1 complex subunit 2), and *complex V* (ATP synthase subunit g, ATP synthase subunit beta). Other important proteins implicated in the mitochondrial oxidative phosphorylation (ADP/ATP translocase 1 protein), ATP production by the Krebs cycle (isocitrate dehydrogenase [NADP] and 2-oxoglutarate dehydrogenase), and the fatty acid metabolism (delta(3,5)-delta(2,4)-dienoyl-CoA isomerase) were also significantly decreased.

Conversely, we observed an important increase in proteins involved in the pyruvate metabolism (isoform 2 of lactoylglutathione lyase), the gluconeogenesis (fructose-1,6-bisphosphatase isozyme 2), the ATPase (isoform SERCA1A of sarcoplasmic/endoplasmic reticulum calcium ATPase 1), and the lipid metabolism (Apolipoprotein-A1). Other increased protein expressions were related to the cytoskeletal assembly (PDZ and LIM domain protein 3), as well as a protein from the RAF kinase

Table 2 Differently expressed proteins in PAH skeletal muscle compared to healthy controls as determined by iTRAQ

Uniprot code	NCBI code	Protein	Function	Ratio PAH/Ctrl	p value
Downregulated proteins					
Q16795	NDUFA9	NADH dehydrogenase [ubiquinone] 1 alpha subcomplex subunit 9. mitochondrial	Oxidative phosphorylation. complex I. mitochondrial metabolism.	0.71	0.004
P22695	UQCRC2	Cytochrome b-c1 complex subunit 2. Mitochondrial	Oxidative phosphorylation. complex III. mitochondrial metabolism.	0.74	0.006
P31930	UQCRC1	Cytochrome b-c1 complex subunit 1. Mitochondrial	Oxidative phosphorylation. complex III. mitochondrial metabolism.	0.74	0.001
O75964	ATP5L	ATP synthase subunit g. mitochondrial	Oxidative phosphorylation. complex V. mitochondrial metabolism.	0.70	0.01
P06576	ATP5B	ATP synthase subunit beta. mitochondrial	Oxidative phosphorylation. complex V. mitochondrial metabolism.	0.74	<0.0001
P48735	IDH2	Isocitrate dehydrogenase [NADP]. mitochondrial	Krebs cycle. mitochondrial metabolism.	0.74	0.05
Q02218	OGDH	2-oxoglutarate dehydrogenase. mitochondrial	Krebs cycle. mitochondrial metabolism.	0.77	0.002
P12235	SLC25A4	ADP/ATP translocase 1	ADP/ATP translocase	0.65	<0.0001
Q13011	ECH1	Delta (3.5)-Delta(2.4)-dienoyl-CoA isomerase. mitochondrial	Fatty acid metabolism	0.57	0.02
Upregulated proteins					
Q04760-2	GLO1	Isoform 2 of Lactoylglutathione lyase	Pyruvate Metabolism	1.39	0.03
O00757	FBP2	Fructose-1.6-bisphosphatase isozyme 2	Gluconeogenesis	1.33	0.0006
O14983-2	ATP2A1	Isoform SERCA1A of Sarcoplasmic/endoplasmic reticulum calcium ATPase 1	ATPase	1.33	0.0007
P30086	PEBP1	Phosphatidylethanolamine-binding protein 1	RAF kinase inhibitor protein	1.29	0.007
Q9UKX2	MYH1	Myosin-1	Motor protein	2.42	<0.0001
Q9UKX2	MYH2	Myosin-2	Motor protein	1.26	<0.0001
P12883	MYH7	Myosin-7	Motor protein	1.26	<0.0001
Q96A32	MYLPF	Myosin regulatory light chain 2. skeletal muscle isoform	Motor protein	1.58	0.03
P02647	APOA1	Apolipoprotein A-I	Lipid metabolism	1.29	<0.0001
Q53GG5	PDLIM3	PDZ and LIM domain protein 3	Cytoskeletal assembly	1.58	<0.0001

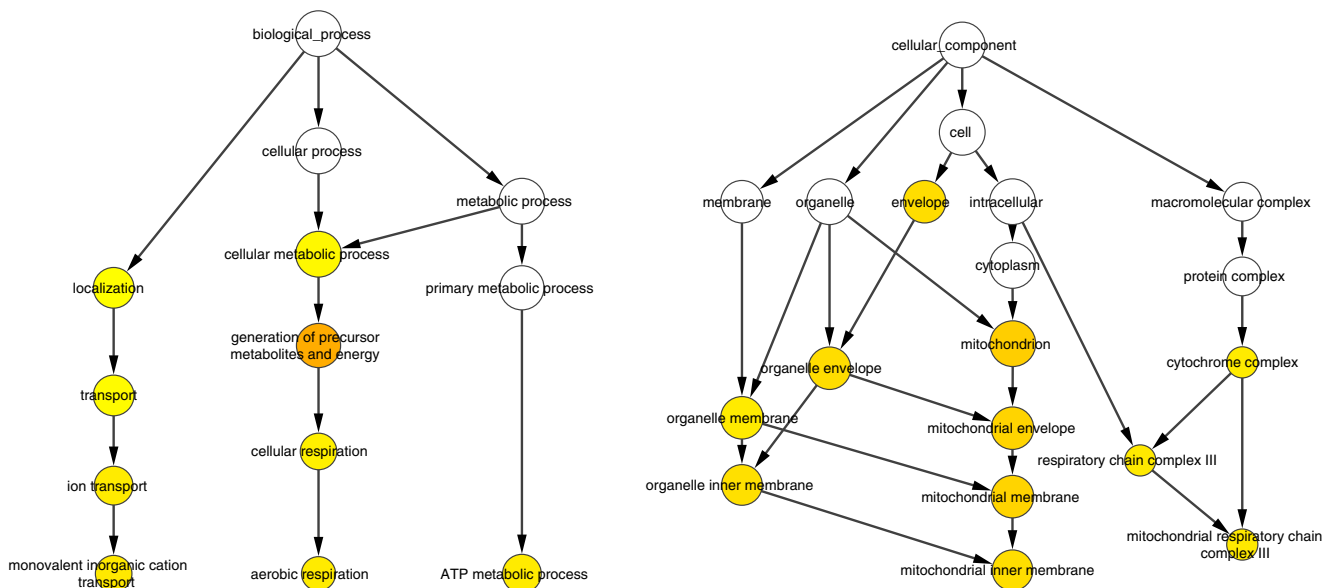


Fig. 2 Mitochondrial metabolism is impaired in PAH skeletal muscle. Bioinformatics analyses from Gene Ontology enrichment displayed (*left*) a downregulation of protein complex of oxidative metabolism and (*right*)

mitochondrial integrity. Note that color intensity is proportional to fold of change of the ratio

inhibitor complex (phosphatidylethanolamine-binding protein 1) and motor proteins (myosin-1, myosin-2, myosin-7, and myosin regulatory light chain 2 skeletal muscle isoform). These findings suggest an altered proteomic signature in the peripheral skeletal muscles of PAH patients and, more specifically, an important imbalance and defect in the mitochondrial oxidative phosphorylation and metabolism.

Abnormal proteomic signature impairs the metabolic pathways in PAH skeletal muscles

Based on this PAH skeletal muscle proteomic profile, we subsequently identified using bioinformatics tools the metabolic pathways that were impaired in skeletal muscles of PAH patients. Using Gene Ontology (GO) enrichment, we first identified protein complexes that are significantly

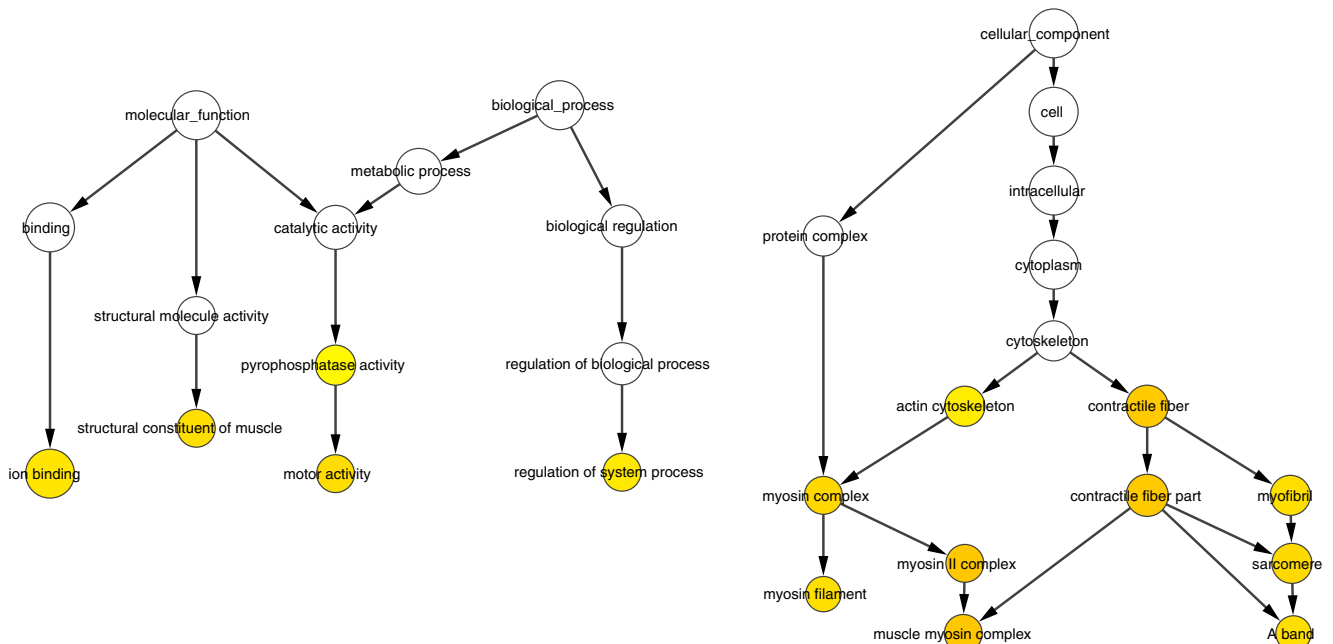


Fig. 3 Skeletal muscle integrity and contractibility are impaired in PAH. Bioinformatics analyses from Gene Ontology enrichment displayed (*left*) an upregulation of protein complex of skeletal muscle integrity and

(*right*) skeletal muscle contractibility. Note that color intensity is proportional to fold of change of the ratio

downregulated and upregulated (Tables E1 and E2, respectively) to characterize which biological networks are impaired in PAH skeletal muscle. Interestingly, GO enrichment showed downregulation of the metabolism processes with impairment of the aerobic respiration, the electron and proton transfer molecules, and the ATP metabolic process (Fig. 2, left). It also showed a downregulation of mitochondrial functions particularly concerning the proton transfer molecules of the complex III (Fig. 2, right). Furthermore, GO enrichment showed upregulation of protein complexes implicated in skeletal muscle integrity (structural constituent of muscle, motor activity, regulation of system process) (Fig. 3, left) and contractile process (myosin filament, muscle myosin complex contractile fiber part) (Fig. 3, right). Both downregulated and upregulated specific pathways were parts of a broad bioinformatic analysis from GO enrichment (Figs. E3 and E4). Thus, our results confirmed strong impairments in the proteomic signature of PAH patients, affecting the aerobic mitochondrial metabolism and the skeletal muscle integrity and contractile function of PAH patients.

Metabolic switch in quadriceps biopsies of PAH patients

Next, we aimed to confirm proteomic analyses experimentally, focusing on the metabolic impairments of PAH muscles. First, PAH patients displayed slightly lower fiber type I proportion and higher fiber type II proportion (Fig. 4a, b). We then quantified the oxidative and anaerobic glycolytic metabolism using PDH and LDH assays, respectively. In accordance with the proteomic signature, PAH skeletal muscle exhibited a significant increase of LDH (Fig. 4d) and decrease of PDH activity (Fig. 4e). These results showed a metabolic switch level from oxidative metabolism to anaerobic glycolysis, therefore suggesting mitochondrial impairment.

The amount of mitochondria is decreased in PAH

Electronic microscopy analysis made on *vastus lateralis* specimens of PAH patients and matched controls revealed abnormal mitochondrial anatomy (Fig. 5a). At growth $\times 50,000$ – $60,000$, we observed in skeletal muscles of PAH patients a global impairment of skeletal muscle structure and organization, including important lipid inclusion and decreased mitochondrial density. Then, to focus on mitochondria, we performed microscopy structure analysis at growth $\times 50,000$ – $60,000$ and observed enlarged and misshaped mitochondria with a loss of mitochondrial crest. We then confirmed our microscopic observations and quantified mitochondria density using validated techniques of CS assay (Fig. 5b) and Western blot against TFAM (Fig. 5c). As expected, we observed a decrease of CS activity and

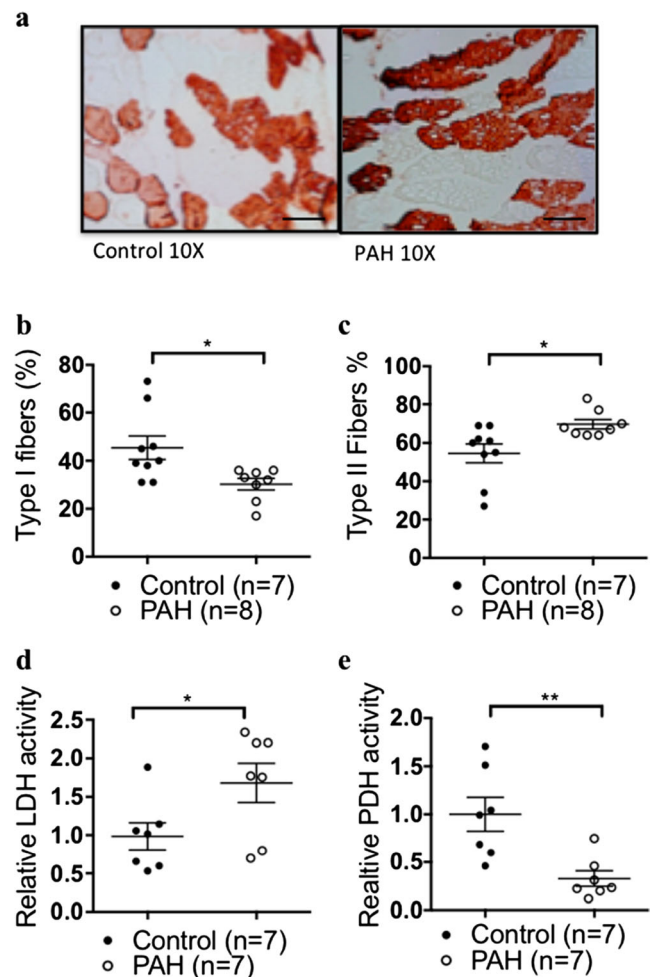


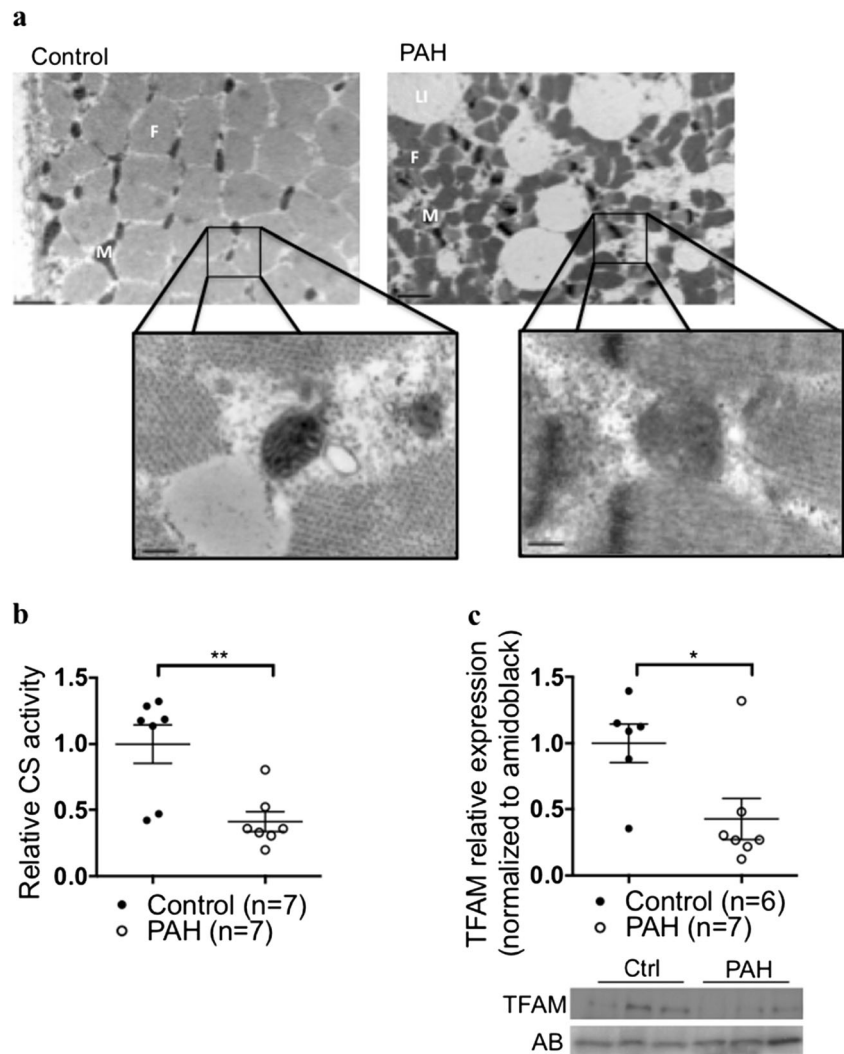
Fig. 4 Metabolic switch observed in PAH skeletal muscle. **a** Immunohistochemistry fiber typing (type I non-stained; type II stained; 10 $\times$), and **b**, **c** showed significant decrease of type I fibers and increase of type II fibers in skeletal muscle of PAH ($n=8$) compared to matched control ($n=8$). Enzymatic activity measured in human biopsies showed significant increased of LDH activity (**d**) and decreased of PDH activity (**e**) in skeletal muscle of seven PAH compared to six matched controls. Scale bar value 100 μ m. Values are expressed as mean $\pm$ SEM; * $p<0.05$; ** $p<0.01$

TFAM protein expression in PAH. Altogether, these results suggest strong impairment in integrity and density of the mitochondria in PAH skeletal muscles (Fig. 6), confirming previous finding in pulmonary artery smooth muscle cells [19–21].

Oxidative impairment correlates with exercise tolerance in PAH

Among PAH patients, exercise capacity positively correlated with skeletal muscle oxidative capacity (Table 3), revealing a possible implication of oxidative impairments in exercise intolerance in PAH. Exploratory analysis also suggested a

Fig. 5 Total amount of mitochondria is decreased in PAH skeletal muscle. **a** Electronic microscopy analysis of skeletal muscle specimens of PAH patients and healthy controls showed (1) muscle fiber organization impairment, (2) lipid inclusion, and (3) decrease of mitochondria number (at growth $\times 4000$ – 6000). Increased growth ($\times 50,000$ – $60,000$) showed defective PAH mitochondria which had burst and displayed impairment of mitochondrial crest (*m* mitochondria, *F* fibers, *LI* lipid inclusion). **b**, **c** Both citrate synthase (CS) activity (seven PAH, six healthy control) and protein expression of TFAM (seven PAH, five healthy control) were significantly decreased in PAH. **a** Scale bar value are $1\ \mu\text{m}$ and $0.2\ \mu\text{m}$ for both Ctrl and PAH. Values are expressed as mean $\pm$ SEM; * $p < 0.05$; ** $p < 0.01$



positive correlation between skeletal muscle oxidative capacity and quadriceps strength (Table 3).

Discussion

In the present study, the whole proteomic profile of PAH skeletal muscle was performed using the latest and state-of-the-art iTRAQ technology, allowing better protein quantification than classical immunoblots [22]. Thus, our iTRAQ analysis displayed 9 downregulated proteins and 10 upregulated proteins, pointing to defects in the mitochondrial oxidative phosphorylation and metabolism, muscle integrity, and contractile function. Driven by advanced proteomic analysis in silico, we identified unique molecular network architectures that orchestrate mitochondrial aerobic and metabolic impairments and structural abnormalities. We also experimentally confirmed a metabolic switch with a decreased activity of the oxidative metabolism enzyme (PDH) and an increased

activity of anaerobic glycolysis enzyme (LDH), as previously seen in pulmonary circulation [23] and RV [24]. Finally, we showed that mitochondrial structure (electronic microscopy) and number (CS activity and TFAM) were drastically different compared to healthy controls. Given that the ability to sustain a steady-state level of muscular work is a function of the volume and density of mitochondria [25, 26], these alterations in mitochondrial function within skeletal muscles of patients, combined with impaired central hemodynamics [27, 28] and angiogenesis [6], are likely to contribute to PAH-related declines in whole-body peak oxygen uptake [29].

More recently, our laboratory documented that a miR-126 loss of expression in the peripheral muscle endothelial cells leads to an important decrease of microvessel density in the skeletal muscle of PAH patients [6]. This loss of skeletal muscle microcirculation density might lead to a lower supply in oxygen, nutrients, and growth factors to exercising muscles. Interestingly, mitochondrial metabolism and angiogenesis are closely linked as vascular branching signals regulate mitochondrial biogenesis [30]. Conversely, miR-126

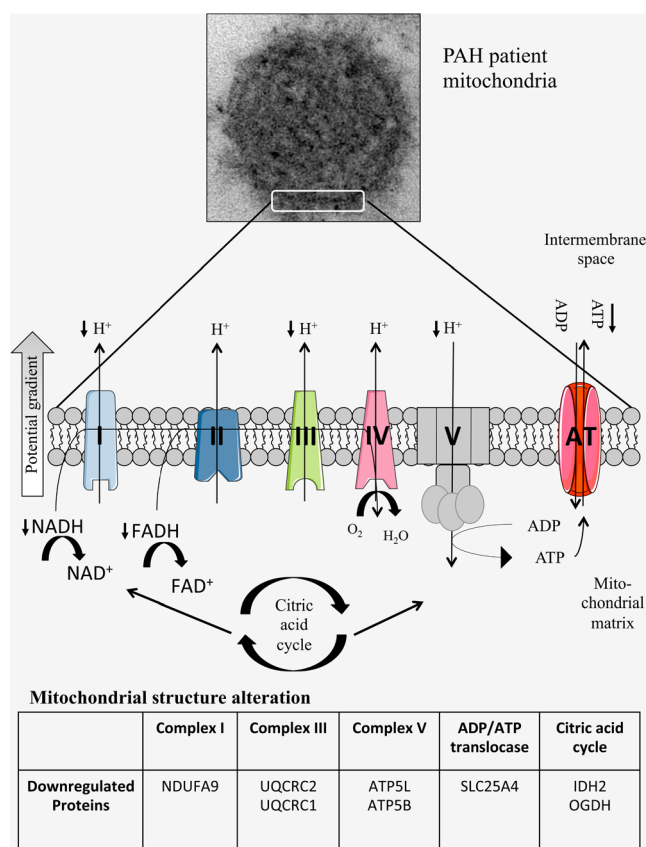


Fig. 6 Summary of the outlining proteomic changes in skeletal muscle mitochondria of PAH patients. PAH patient mitochondria are displaying abnormal structure (electron microscopy; $\times 50,000$ – $60,000$). Complex I to V form the electron transport chain. Structure alterations in the electron transport chain might impair H^+ transport and later decreasing ATP formation, hence favoring metabolic switch to glycolytic ATP formation for energy supply. Alteration in the citric acid cycle might lead to lower formation of NADH, FADH, and eventually ATP formation, also contributing to the metabolic switch. *Complex I* NADH dehydrogenase, *complex II* succinate dehydrogenase, *complex III* cytochrome $b-c_1$, *complex IV* cytochrome c oxidase, *complex V* ATP synthase, *AT* adenosine nucleotide translocase, *NADH* nicotinamide adenine dinucleotide, *NAD⁺* reduced nicotinamide adenine dinucleotide, *FADH* flavin adenine dinucleotide, *FAD⁺* reduced flavin adenine dinucleotide. Proteins displayed in the table are identified using the NCBI code (refer to Table 2)

expression is regulated in part by mitochondria destabilization stimuli, reducing mitochondrial respiration and increasing glycolysis production of ATP in malignant mesothelioma cells [31]. Therefore, the metabolic/mitochondrial impairment in PAH might contribute to the recently described skeletal muscle microcirculation loss, and they might also jointly act on decreasing exercise capacity in PAH.

The reason why relatively identical mitochondrial abnormalities occur simultaneously in PAH lungs, RV, and skeletal muscles is of major interest. In accordance with our results, past work focusing on proteomic analysis of plasma samples and lung tissues from PAH patients has identified differently expressed proteins with roles in cell growth, proliferation, and

Table 3 Correlation between oxidative and mitochondrial impairments and vastus lateralis morphology and exercise capacity

	Pearson's <i>r</i>	<i>p</i> value
VO₂ max (ml min⁻¹ kg⁻¹)		
PDH	0.70	0.011
LDH	−0.65	0.022
CS	0.84	0.001
TFAM	0.68	0.021
Voluntary strength (kg)		
PDH	0.60	0.052
LDH	−0.50	0.121
CS	0.77	0.006
TFAM	0.37	0.289
Endurance (N m)		
PDH	0.55	0.076
LDH	−0.28	0.403
CS	0.51	0.108
TFAM	0.50	0.142

PDH pyruvate dehydrogenase, LDH lactate dehydrogenase, CS citrate synthase, TFAM transcription factor A of the mitochondria

cell metabolism [32, 33]. Furthermore, several evidences showed that mitochondria contribute to this metabolic dysfunction and proliferation/apoptosis imbalance of pulmonary artery smooth muscle cells. Indeed, mitochondria not only are responsible for producing energy but also control cell cycle, metabolism, development, and cell death [34]. Similar mitochondrial remodeling and metabolic switch toward glycolysis also contribute to RV failure in PAH [35]. Interestingly, skeletal muscle mitochondrial dysfunction was recently described in monocrotaline-induced PAH rats [13]. Skeletal muscle abnormalities even preceded RV impairment in this rat model, suggesting that mitochondrial impairment at the skeletal muscle level could be a potential biomarker for RV failure in PAH. Thus, our findings suggest that the PAH mitochondrial defects described in the lungs, RV, and skeletal muscle represent a more systemic phenomenon than initially thought, as recently suggested by Sutendra and Michelakis [7].

Systemic inflammation could represent a common mechanism accounting for these systemic mitochondrial defects. Plasma levels of IL-1 β and IL-6, both potent proinflammatory cytokines, are elevated in PAH [36]. Bertero et al. recently documented that these inflammatory cytokines triggered the upregulation of miR-130/301 complex in pulmonary artery smooth muscle cells, endothelial cells, and in plasma of PAH patients, which in turn downregulated the peroxisome proliferator-activator receptor γ (PPAR γ) [37]. PPAR γ plays a critical role in the molecular ethology of PAH, as its downregulation has been shown to be a key regulator in the upregulated glucose metabolism and in the abnormal mitochondrial biogenesis in pulmonary arterial smooth muscle cells [38].

Taken together, increased systemic inflammation might up-regulate the miR-130/301 complex in muscular cells, which downregulates PPAR γ and negatively impacts on systemic mitochondrial functions by altering their proteomic formations, ultimately leading to a progressive lowering of oxidative function.

The molecular action of PPAR γ also impacts insulin sensitization [39]. Interestingly, PPAR γ -mediated insulin resistance is increasingly recognized in PAH [40–42]. Muscle ectopic deposition of lipids, as observed in our PAH patients on electronic microscopy (Fig. 5a), is also a common feature of the metabolic syndrome leading to the pathogenic development of insulin resistance and type 2 diabetes [43]. While skeletal mitochondrial proteins affect insulin responsiveness and glucose uptake in skeletal muscles, insulin also emerged as an important regulator of mitochondrial biogenesis [44]. It has been hypothesized that this signaling pathway may be blunted in insulin-resistant individuals and that decreased mitochondrial content and function may be a consequence of impaired insulin signaling rather than the opposite [45]. Taken altogether, systemic inflammation and insulin resistance might be etiologically related to skeletal muscle mitochondrial impairment, which in turns downregulates miR-126 expression resulting in a lower angiogenesis and exercise intolerance.

Proteomic profile also showed an upregulation of one protein entering the structure of an enzyme implicated in the fructose 1,6-biphosphate metabolisms (the fructose 1,6-biphosphate isozyme 2). This enzyme is linked to a concomitant increase in another protein entering in the formation of the lactoylglutathione lyase (lactoylglutathione lyase isoform 2), which is a by-product of the neutralization of the methylglyoxal, a highly reactive and essential compound derived from the glycolysis pathway [46]. Interestingly, recent data indicated that increased intravenous and intramuscular concentrations of methylglyoxal induces PAH in an animal model [47]. Its vasotoxicity is associated with increased inflammation, reactive oxygen species production, DNA damage, and vascular endothelial dysfunction, all proved impacting mitochondrial health and being implicated in the pathophysiology of PAH [48, 49]. Therefore, the increased expression of methylglyoxal and its metabolites represent an attractive hypothesis to explain mitochondrial dysfunction and warrants further investigation.

Other cardiopulmonary diseases like chronic obstructive pulmonary disease (COPD) are known for broad morphological and structural alteration of the skeletal muscle [50]. Reduced CS activity has been correlated with time to muscle fatigue [51], increased arterial lactate level, lower oxidative capacity, and peak exercise capacity [52], independently from lung function impairments [53]. Increased oxidative stress has strong functional implications for the contractile leg function. Hence, the multiple abnormalities observed in COPD begin to

be observed in PAH and might help the clinician to better understand skeletal muscle dysfunction and adapt their treatment strategies in this later disease. For example, cardiopulmonary rehabilitation is of interest as it improves muscle mass and oxygen delivery to mitochondria with reversal of some of the mitochondrial changes observed in COPD [54] and proved its efficiency on skeletal muscle morphology and microcirculation in PAH as well [55, 56].

Our study contains limitations that need to be addressed. First, our results pointed out an upregulation of protein complexes implicated in skeletal muscle integrity and in the contractile function. These results were unexpected knowing that PAH patients display significant decreases in quadriceps voluntary strength, endurance, and morphology [57]. As our results also identified major skeletal muscle mitochondrial defects, the lower level of strength and endurance of patients may originate more from compensative mechanisms to cope with lower levels of energetic supplies, commonly mirrored by decreased daily activities and increased fatigability. The number of patients in our study is relatively small. However, the initial design of the study was to identify a large number of proteins and later identify upregulated and downregulated proteins. Tissue availability later limits morphological and metabolic analysis. Nonetheless, our results are in accordance with the current literature in PAH regarding skeletal muscle dysfunction [5, 6, 57]. Finally, correlations between mitochondrial defects, muscle characteristics, and exercise capacity should not be seen as a causal relationship but rather as potent contributors to exercise intolerance in relation with other mechanisms already described in the literature.

Conclusion

In conclusion, we demonstrated for the first time that human skeletal muscle in PAH exhibit a specific and unique proteomic signature when compared to healthy controls, with a metabolic switch from an oxidative mitochondrial metabolism to a glycolytic metabolism and a decrease in mitochondrial density. As similar mitochondrial abnormalities are recognized features of PAH pulmonary artery smooth muscle cells and RV, our data suggests that PAH is more than a disorder of the heart and lungs but a systemic disease with multiple organ impairments. These preliminary observations warrant more investigation to fully understand the pathophysiological basis of these skeletal muscle mitochondrial and metabolic defects in PAH.

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